Num Anal Homework-4

1. A small community has a population of 400, of which, at time 0, 201 are infected with a disease and thirteen are vaccinated against the disease. The infected are well mixed within the community. Write down an appropriate disease model. If the rate of infection is 0.001 and the rate of recovery is 0.1, show that the disease will grow. When the disease epidemic is at its highest point, how many people are infected? At that time, how many susceptible persons are there? Finally, when the disease dies out, are there any susceptible persons left? Give your answer and show how you arrived at the answer.

2. Find the solution P_n using the given initial value P_0 of the following models.

   Model 1: $P_{n+1} = 7 P_n / (3 P_n + 1)$, initial value $P_0 = 1$.

   Model 2: $P_{n+1} = (P_n - 5)^2 + 5$, initial value $P_0 = 6$.

Lazizbek Sadullaev. MATH 548. HW #4.

Problem 1.

(i) If the infected are well mixed within the community write down appropriate <u>disease model</u>:

$$(1)\begin{cases} \dfrac{dS}{dt} = -\alpha S I \\ \dfrac{dI}{dt} = \alpha S I - \gamma I \\ \dfrac{dR}{dt} = \gamma I. \end{cases}$$

where

$\alpha = 0.001$ = infection rate

$\gamma = 0.1$ = recovery rate

(2) $S + I + R = N$ = total # of pop-n.

at $t=0$:

(3) $I_0 = I(t=0) = 201$, $S_0 = 400 - (201+13) = 186$ &

$R_0 = R(t=0) = 13$. (given)

The first 2 eq-ns of (1):

$$(4)\begin{cases} \dfrac{dI}{dt} = \alpha S I - \gamma I \\ \dfrac{dS}{dt} = -\alpha S I \end{cases}$$

$\div$ Dividing these:

(5) $\frac{dI}{dS} = -1 + \frac{\gamma I}{\alpha S I} = \frac{\gamma}{\alpha} \cdot \frac{1}{S} - 1. \quad \Big\} \cdot dS$

$$\int dI = \int \left(\frac{\gamma}{\alpha} \cdot \frac{1}{S} - 1 \right) dS + C. \quad \Rightarrow$$

(6) $I(S) = \frac{\gamma}{\alpha} \ln(S) - S + C \quad ; \quad C = ?$

at $t = 0$.

(7) $I(S_0) = I_0 = -S_0 + \frac{\gamma}{\alpha} \cdot \ln(S_0) + C. \quad \Rightarrow$

(8) $C = (I_0 + S_0) - \frac{\gamma}{\alpha} \ln(S_0)$

Pluggiy (8) to (6):

(9) $I(S) = \frac{\gamma}{\alpha} \ln(S) - S + (I_0 - S_0) - \frac{\gamma}{\alpha} \ln(S_0) \quad \Rightarrow$

(10) $I(S) = (N_0 - R_0) - S + \frac{\gamma}{\alpha} \ln\left(\frac{S}{S_0}\right)$ General Model.

(11) $\begin{cases} \text{For our problem, } \alpha = 0.001, \ \gamma = 0.1 \ \& \\ I_0 = 201, \ S_0 = 186, \ R_0 = 13 \end{cases}$

(10) becomes $I(S) = (400 - 13) - S + \frac{0.1}{0.001} \ln\left(\frac{S}{186}\right) \quad \Rightarrow$

(12) $\boxed{I(S) = 397 - S + 100 \cdot \ln\left(\frac{S}{186}\right).}$

(ii) Show that the disease will grow given $\alpha = 0.001$ & $\gamma = 0.1$ and $S_0 = 186$ at $t = 0$.

By (1)

(13) $\frac{dI}{dt} = \alpha S I - \gamma I > 0. \Rightarrow \boxed{I'(t) > 0 \text{ to grow}}$

$$\Rightarrow \overbrace{\underbrace{\gamma}_{\substack{\oplus \\ \text{recov. rate}>0}} \cdot \underbrace{I}_{\substack{\oplus \\ \text{\# of people}}}}^{\oplus\, >0} \left(S \cdot \frac{\alpha}{\gamma} - 1\right) > 0 \Rightarrow \frac{S \cdot \alpha}{\gamma} > 1$$

For disease to grow given time t is:

(14) $\frac{S(t) \cdot \alpha}{\gamma} > 1$ = "Reproduction number"

at $t = 0$ $S(0) = S_0 = 186$, so

(15) $\frac{S(t) \cdot \alpha}{\gamma} = \frac{186 \cdot 0.001}{0.1} = 1.86 > 1 \Rightarrow$

So the disease actually grow!

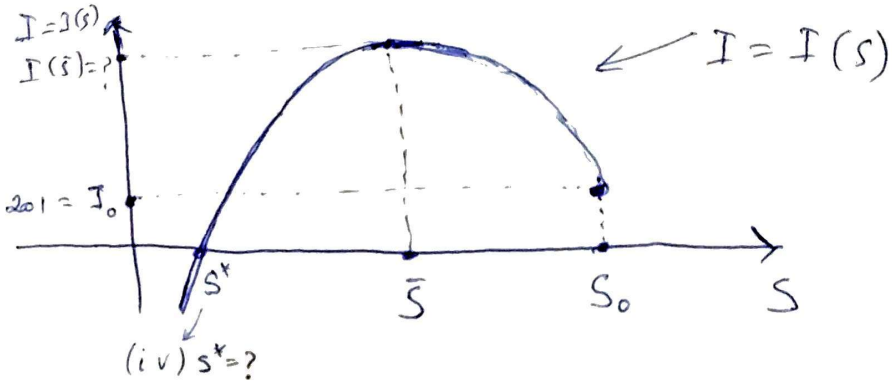
Indeed, let me graph the function (12) on a separate page:

(iii) when the disease is the highest, how many people are infected & susceptible ?

(12) $I(S) = 397 - S + 100 \cdot \ln\left(\frac{S}{186}\right)$

So the graph of this $I(S)$ is similar to this:

(16)



By question (iii), we need to find $\bar{S} = ?$ & $I(\bar{S}) = \bar{I} = ?$

For this, we need to find $\frac{d}{dS} I(S) = 0 \Rightarrow \bar{S} = ?$

(17) $I'(S) = \frac{d}{dS} I(S) = \frac{d}{dS}\left(397 - S + 100 \cdot \ln\left(\frac{S}{186}\right)\right) =$

$= -1 + 100 \cdot \frac{1}{\frac{S}{186}} \cdot \frac{1}{186} =$

$= -1 + \frac{100}{S}$

(18) $I'(\bar{S}) = 0 \Rightarrow -1 + \frac{100}{\bar{S}} = 0 \Rightarrow$

(19) $\bar{S} = 100 \Rightarrow I(\bar{S}) = 397 - 100 + 100 \cdot \ln\left(\frac{100}{186}\right) = 234{,}9$

(20) $\bar{S} = 100$ & $\bar{I} = I(\bar{S}) \approx 235.$

math548_hw4_Lazizbek

February 9, 2024

1 Lazizbek Sadullaev, Math 548, HW #4

Problem 1

A small community has a population of 400, of which, at time 0, 201 are infected with a disease and thirteen are vaccinated against the disease. The infected are well mixed within the community. Write down an appropriate disease model. If the rate of infection is 0.001 and the rate of recovery is 0.1, show that the disease will grow. When the disease epidemic is at its highest point, how many people are infected? At that time, how many susceptible persons are there? Finally, when the disease dies out, are there any susceptible persons left? Give your answer and show how you arrived at the answer.

The model based on my previous calculations is as follows:

I(S) = 397 - S + 100*ln(S/186)

When the disease dies out, are there any susceptible persons left? I am going to solve I(S*) = 0 to answer the question above. For that, since we have Newton failing point at I'(S)=0, I am going to use the **Bisection** method

[18]:
```
import numpy as np
import math

from scipy.optimize import fsolve
a = 0.001 # since ln(S/186) has a discontinuity at s = 0, is s* < 1,
# then I know there's no susceptible so no worry about that in case of the␣
↪root very close to 0.
b = 100 # I am starting the right side from max point, s bar.
solution_int = [a, b]

delta = 0.0000001
epsilon = 0.0000001
count = 0
# |x-xn|<epsilon, |x-xn|<bn-an, if bn-an<epsilon, i.e. (b0-a0)/2**n<epsilon
# (b0-a0)/epsilon < 2**n
# math.log((b0-a0)/epsilon, 2) < n, So
print(f"Please enter M as bigger than {math.ceil(math.log((b-a)/epsilon, 2))}␣
↪")
M = int(input("M = "))
```

```
u = 397 - a + 100*math.log(a/186)
v= 397 - b + 100*math.log(b/186)
e = b - a
print("a = ", a)
print("f(a) = ", u)
print("f(b) = ", v)

if np.sign(u) == np.sign(v):
    print("There's no simple solution because the signs of the function at␣
↪the boundries are eqaul")
    count = count + 1
else:
    for k in range(1, M):
        count = count + 1
        e = e/2
        c = a + e
        w = 397 - c + 100*math.log(c/186)
        if abs(e) < delta or abs(w)<epsilon:
            break
        if np.sign(w) != np.sign(u):
            b = c
            v = w
        else:
            a = c
            u = w

    f = lambda x: 397 - x + 100*math.log(x/186)
    print("number of iterations k = ", k)
    print(f"Using Python's fsolve function, fsolve(f, {solution_int}) = ", ␣
↪*fsolve(f, 0.01))
    print("middle of the interval , and in turn approximate solution c = ", c)
    print("funtion value in the middle w = ", w)
    print("halving the interval e = ", e)
```

```
Please enter M as bigger than 30
M = 36
a =  0.001
f(a) =  -816.3511952695337
f(b) =  234.94235122748898
number of iterations k =  30
Using Python's fsolve function, fsolve(f, [0.001, 100]) =  3.6406157567362083
middle of the interval , and in turn approximate solution c =  3.640615717840381
funtion value in the middle w =  -1.0294901926499733e-06
halving the interval e =  9.313132613897323e-08
```

[ ]:

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Problem 2:

Find the solution P_n using the given initial value P_0 of the ↓ models. (i):

(21). $P_{n+1} = \dfrac{7P_n}{3P_n+1}$, $P_0 = 1$.

(22) $(P_{n+1})^{-1} = \left(\dfrac{7P_n}{3P_n+1}\right)^{-1} \Rightarrow \dfrac{1}{P_{n+1}} = \dfrac{3P_n+1}{7P_n} \Rightarrow$

$$\frac{1}{P_{n+1}} = \frac{3P_n}{7P_n} + \frac{1}{7P_n} = \left(\frac{3}{7}\right) + \left(\frac{1}{7}\right)\cdot\frac{1}{P_n} \Rightarrow$$

(23) $$\frac{1}{P_{n+1}} = \left(\frac{3}{7}\right) + \left(\frac{1}{7}\right)\cdot\frac{1}{P_n}$$

Now let $y_n = \dfrac{1}{P_n}$ as a new variable $\Rightarrow$

(24) $y_n = \dfrac{1}{P_n} \Rightarrow y_{n+1} = \dfrac{1}{P_{n+1}}$

Plugging (24) to (23):

$$y_{n+1} = \left(\frac{3}{7}\right) + \left(\frac{1}{7}\right)\cdot y_n \qquad \text{or}$$

(25) $$y_{n+1} - \left(\frac{1}{7}\right)\cdot y_n = \frac{3}{7}$$

To solve this (25) difference equation, like in differential eq., we first solve the homogeneous part & then find a particular solution for (25) & then add these 2.

(26) $y_{comp.} = \overset{a)}{y_{hom.}} + \overset{b)}{y_{part.}}$

Problem 2:

Find the solution P_n using the given initial value P_0 of the models: (i):

(21). $P_{n+1} = \dfrac{7P_n}{3P_n+1}$, $P_0 = 1$.

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$$\frac{1}{P_{n+1}} = \frac{3P_n}{7P_n} + \frac{1}{7P_n} = \left(\frac{3}{7}\right) + \left(\frac{1}{7}\right)\cdot\frac{1}{P_n} \Rightarrow$$

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Now let $y_n = \dfrac{1}{P_n}$ as a new variable $\Rightarrow$

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(25) $$y_{n+1} - \left(\frac{1}{7}\right)\cdot y_n = \frac{3}{7}$$

To solve this (25) difference equation, like in differential eq-ns, we first solve the homogeneous part & then find a particular solution for (25) & then add these 2.

(26) $y_{comp.} = \overset{a)}{y_{hom.}} + \overset{b)}{y_{part.}}$

a) Homogenous part:

(27) $y_{n+1} - \left(\frac{1}{7}\right) \cdot y_n = 0.$

$$\lambda - \frac{1}{7} = 0 \Rightarrow \lambda = \frac{1}{7} \Rightarrow$$

(28) $y_h = A \cdot \left(\frac{1}{7}\right)^n$

b) Particular (non homogeneous part):

(29) $y_{n+1} - \left(\frac{1}{7}\right) y_n = \frac{3}{7}.$

If this (29) is approaching when at steady state

(30) $y_{n+1} = y_n = y^* \Rightarrow \{y_n\} \to y^*$ because

if I write

(31) $y_{n+1} = \left(\frac{1}{7}\right) y_n + \frac{3}{7}$ & by fixed point

iteration $g(x)$:

(32) $x = \underbrace{\left(\frac{1}{7}\right) x + \frac{3}{7}}_{g(x)} \Rightarrow$ so if

(33) $g(x) = \left(\frac{1}{7}\right) x + \frac{3}{7}$ $\quad |g'(x)| = \left|\frac{1}{7}\right| < 1 \ \forall x.$

so (29) has a steady state <u>regardless of y_n's!</u>

So, at this steady state $\{y_n\} \to y^* \Rightarrow y_{n+1} = y_n = y^*$:

Plugging this to (29), we get a <u>particular</u> solution:

(34) $y^* - \left(\frac{1}{7}\right) y^* = y^* \Rightarrow \quad y^* = \frac{1}{2}$, so

(35) $y_p = \frac{1}{2}.$

Hence, by (26) $y_{complete} = y_{hom.} + y_{part}$,
the solution for (25) is (28) + (35):

(36) $y_n = A \cdot \left(\frac{1}{7}\right)^n + \frac{1}{2}.$

Also recall (24) $y_n = 1/p_n$, so (36) is

(37) $\frac{1}{p_n} = A \cdot \left(\frac{1}{7}\right)^n + \frac{1}{2}.$ at $n=0$, $P_0 = 1$ by (21):

$$\frac{1}{P_0} = A \cdot \left(\frac{1}{7}\right)^0 + \frac{1}{2} \Rightarrow \quad A + \frac{1}{2} = 1 \Rightarrow A = \frac{1}{2}.$$

So, by (37)

$$\frac{1}{P_n} = \left(\frac{1}{2}\right) \cdot \left(\frac{1}{7}\right)^n + \frac{1}{2}. \quad \&$$

(38) $P_n = \left(\frac{1}{2} \cdot \left(\frac{1}{7}\right)^n + \frac{1}{2}\right)^{-1}.$

(ii) $P_{n+1} = (P_n - 5)^2 + 5 \Big\}_{-5}$ $\quad P_0 = 6.$ $\Rightarrow$

(39) $(P_{n+1} - 5) = (P_n - 5)^2 \; \{ \ln$

(40) $\ln(P_{n+1} - 5) = 2 \ln(P_n - 5)$

Let $y_n = \ln(P_n - 5)$, then $\Rightarrow$

(41) $y_n = \ln(P_n - 5) \Rightarrow y_{n+1} = \ln(P_{n+1} - 5)$

(42) $y_{n+1} = 2 y_n \Rightarrow \lambda - 2 = 0 \Rightarrow \lambda = 2.$

(43) $y_n = A \cdot 2^n$ and recall $y_n = \ln(P_n - 5)$ so,

(44) $A \cdot 2^n = \ln(P_n - 5) \Rightarrow \boxed{P_n = 5 + e^{A \cdot 2^n}}$ & at $t = 0$

$P_0 = 5 + e^{A \cdot 2^0} = 6$ by (39).

$e^A = 1 \Rightarrow A = 0.$ $\Rightarrow$

$P_n = 5 + e^{0 \cdot 2^n} = 5 + e^0 = 6. \Rightarrow$

(45) $\boxed{P_n = 6}$